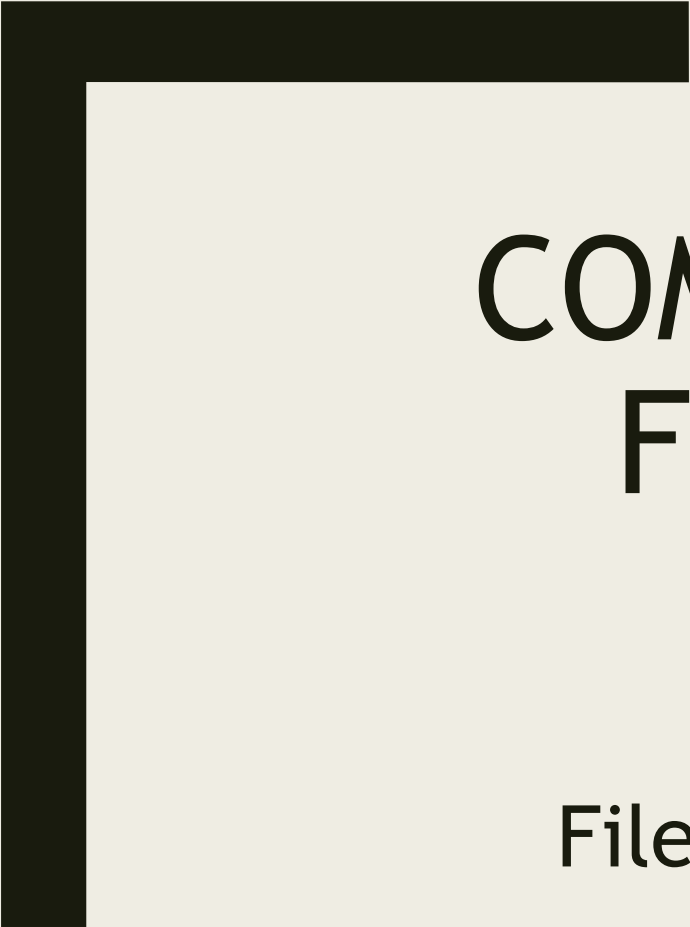
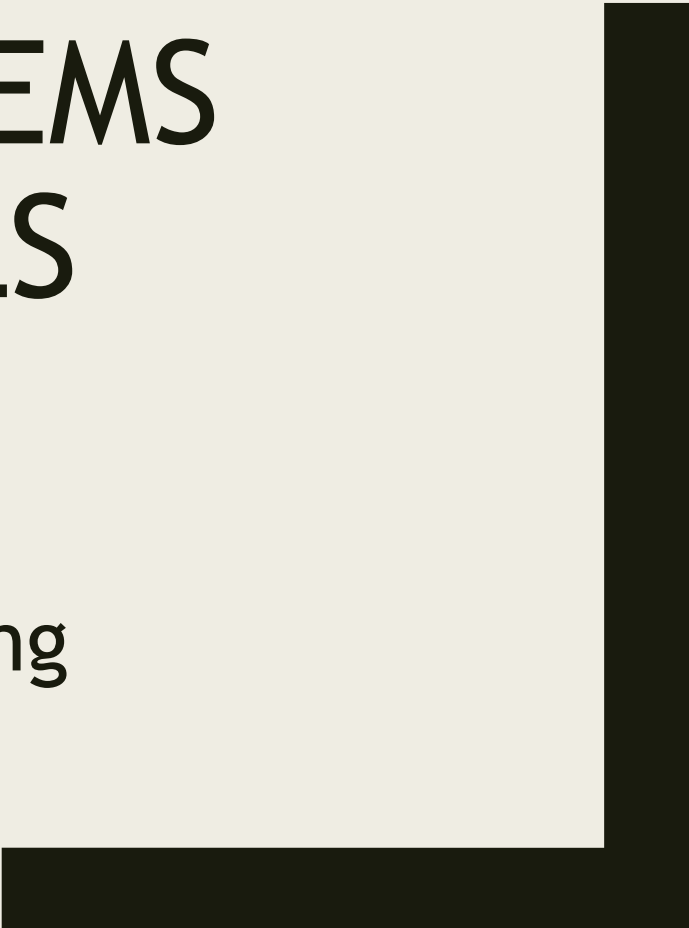




COMPUTER SYSTEMS FUNDAMENTALS (4COSC009C)

File types & Disk scheduling



Contact details:

In File Systems part b:

- FAT File Systems
 - *FAT structure*
 - *FAT Boot Sector*
- Windows volumes
- Unix volumes
- Directories
- Absolute path names
- Relative path names
- File types
- File operations
- Unix file systems
- Disk scheduling

File Types

Types of files

Confusion: Text files vs. Binary Files

■ Text files (ASCII / UNICODE)

- *Bytes of data are organised as characters from respective character sets*

■ Binary files

- *Data in a specific format that requires interpretation.*

■ Text files vs. Binary Files

- *All files are in Binary*
- *Text Files are formatted in chunks of 8 bits or 16 bits*
- *Files in any other format are Binary Files*

File types

- Most files contain a specific types of information
 - *A Java program*
 - *A JPEG image*
 - *A BITMAP image*
 - *An MP3 clip*
- The kind of information is the **file type** tells which application or operation to use
 - *So the File System knows which operations it can do*
 - *Most OS have associations between file types and applications*

list file extensions you use daily and match the applications

File Types *Extensions*

- File names are often separated by a full-stop into 2 parts
 - *Main name*
 - *File extension*
- The **file extension** was used by the OS to identify the type of file
 - *But is not necessarily the actual file type*
- Windows 10 will inspect the file to ascertain the actual file type
 - *Looking at the file header*

Extension	File Type
.txt	Text data file
.mp3 , .au, .wav	Audio file
.gif , .tiff , .jpg	Image file
.doc , .odt	Word processing files

Anatomy of an ASCII File

The screenshot shows the WinHex application window with the file 'textfile.txt' open. The main window displays the hex and ASCII representation of the file's content. A 'Hexadecimal / ANSI ASCII' table is overlaid on the screen, showing the mapping between hexadecimal values and their corresponding ASCII characters.

WinHex - [textfile.txt]

File Edit Search Navigation View Tools Specialist Options Window Help

Case Data

File Edit

textfile.txt

Offset	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00000000	54	68	69	73	20	69	73	20	61	20	74	65	78	74	20	66
00000016	69	6C	65													

This is a text file

[unregistered]

textfile.txt
C:\Users\weingan\Pictures

File size: 19 B
19 bytes

Default Edit Mode
State: original

Undo level: 0
Undo reverses: n/a

Creation time: 17/01/2017
14:48:45

Last write time: 17/01/2017
14:48:45

n/a Size: n/a

Hexadecimal / ANSI ASCII

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
00	NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT	LF	VT	FF	CR	SO	SI
10	DLE	DC1	DC2	DC3	DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS	RS	US
20		!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
30	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
40	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
50	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
60	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
70	p	q	r	s	t	u	v	w	x	y	z	{		}	~	
80			,	f	"	...				%		<				
90		,	"	"	•	-	-	~				>				
A0		i														
B0																
C0	À	Á	Â	Ã	Ä	Å	Æ	Ç	È	É	Ê	Ë	Ì	Í	Î	Ï
D0	Ð	Ñ	Ò	Ó	Ô	Õ	Ö	×	Ø	Ù	Ú	Û	Ü	Ý	Þ	ß
E0	à	á	â	ã	ä	å	æ	ç	è	é	ê	ë	ì	í	î	ï
F0	ð	ñ	ò	ó	ô	õ	ö	÷	ø	ù	ú	û	ü	ý	þ	ÿ

Revisiting BMP files

consist of bytes representing characters from ASCII or unicode.



WinHex - [220px-SNice.bmp]

File Edit Search Navigation View Tools Specialist

220px-SNice.bmp

Offset	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	ANSI	ASCII
00000000	42	4D	56	37	02	00	00	00	00	00	36	00	00	00	28	00	BMf7	6 (
00000016	00	00	DC	00	00	00	DC	00	00	00	01	00	18	00	00	00	ü ü	
00000032	00	00	30	37	02	00	25	16	00	00	25	16	00	00	00	00	07 %	%
00000048	00	00	00	00	00	00	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000064	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000080	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000096	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000112	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000128	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000144	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000160	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000176	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000192	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000208	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000224	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000240	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y
00000256	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	Y	Y

File signature

[unregistered]
220px-SNice.bmp
C:\Users\weingan\Downloads

File size: 142 KB
145,254 bytes

DOS name: 220PX-~1.BMP

Default Edit Mode
State: original

Undo level: 0
Undo reverses: n/a

Creation time: 12/02/2020 10:16:13

Last write time: 12/02/2020

Anatomy of a Binary file (*jpeg*)

File Headers

(JFIF)
JPEG File Interchange
Format

The screenshot displays the WinHex application window for a file named [64px-A_Smiley.jpg]. The main pane shows the hex dump of the file's header. Annotations with arrows point to specific byte sequences:

- Partial File Header:** Points to the first 16 bytes of the hex dump.
- FF D8 = Start of image marker:** Points to the first two bytes (00000000: FF D8 FF E0).
- FF E0 = JFIF marker:** Points to the fourth byte (00000002: FF E0).
- JFIF Identifier:** Points to the sequence of bytes starting at offset 00000004, which includes the JFIF identifier 'JFIF' and the Adobe Glyph Set name 'Adobe Glyph Set'.

The hex dump shows the following data:

Offset	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00000000	FF	D8	FF	E0	00	10	4A	46	49	46	00	01	01	01	00	78
00000016	00	78	00	00	FF	EE	00	0E	41	61	6F	62	65	00	64	00
00000032	00	00	00	00	FF	E2	FF	FF	49	43	43	5F	50	52	4F	46
00000048	49	4C	45	00	00	09	00	08	80	70	41	44	42	45	02	10
00000064	00	00	70	72	74	72	43	4D	59	4B	4C	61	62	20	07	D0
00000080	1A	00	05	00	29	00	35	61	63	73	70	41	50	00	00	00
00000096	00	00	00	41	44	42	45	00	00	00	00	00	00	00	00	00
00000112	00	00	00	00	00	00	01	00	00	F6	D6	00	00	00	00	00
00000128	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000144	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000160	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000176	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000192	00	FC	00	00	00	74	63	70	72	74	00	00	01	70	00	00
00000208	00	2B	77	74	70	74	00	00	01	9C	00	00	00	14	41	32

Page 1 of 2,342 Offset: 0 = 255 Block: Creation time: 17/01/2017 15:22:27 Last write time: 17/01/2017 15:22:27 Size: n/a

File signatures

- There are file signature databases
 - [*Filesignatures.net*](http://filesignatures.net)
- Wikipedia often has high quality listings of the entire file header

File operations

- Create a file
- Delete a file
- Open a file
- Close a file
- Read data from a file
- Write data to a file
- Reposition the current file pointer in a file
- Append data to the end of a file
- Truncate a file
 - *ie. delete all or part of it*
- Rename a file
- Copy a file

file pointer(file handler) is a reference that lets a programme access a file store of the disk . It tracks the current state of access (cross positions , mode such as read or write etc) and is used by the operating system or library to map file input , output request to the actual data

when you read a file the file pointer moves through the file's byte ,some file types have headers , that identify the type the pointer's current position matters when you parse the header vs body

File protection

- Multi-user Systems
- Access control
 - *Controls who can access files*
 - Who can read
 - Who can write
 - Who can execute

Cyber-security triad

prevent unauthorized modifications and unavailability and ensure authorized users can access data

■ Three dimensions of cyber-security:

- *Confidentiality*
- *Integrity*
- *Accessibility*

Confidentiality:

- Preventing access
- Keep the *bad-guy* out

Accessibility:

- Ensure access
- Make sure the *good-guy* can access the data

Integrity:

- Keep control of any changes made to the data
- Who can change it
- Keep track of any changes

File permissions (windows)

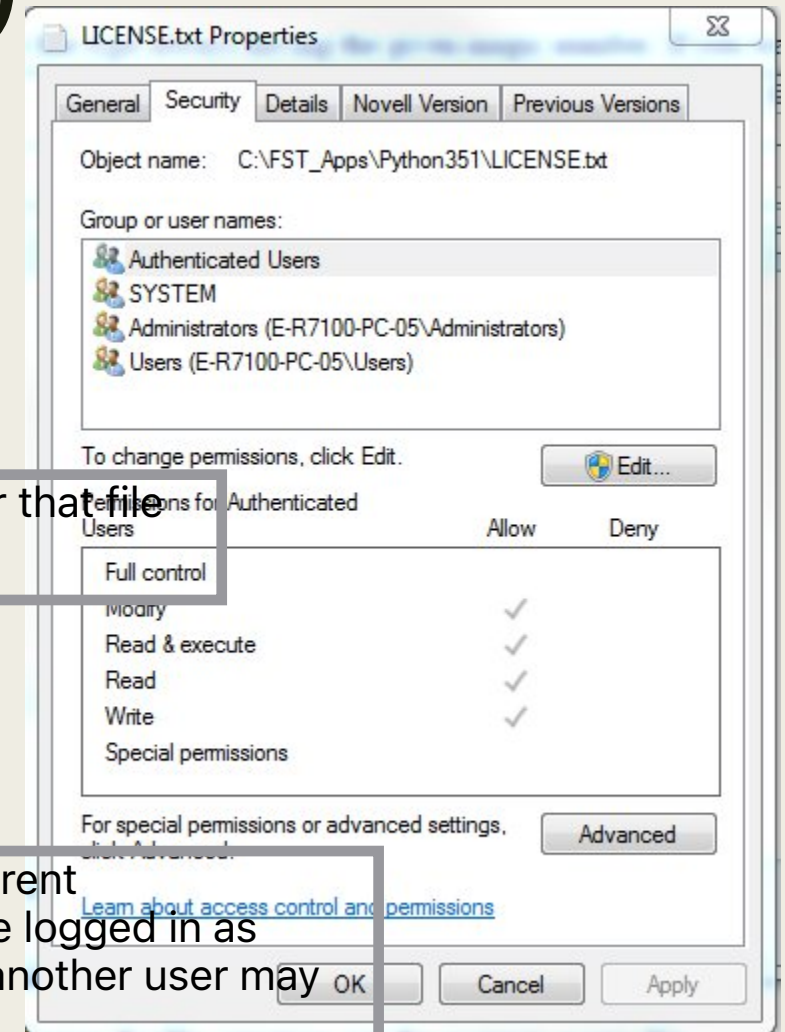
■ NTFS:

- *Access Control Lists (ACL's)*
 - Each file has list of user identities with permissions
- *Explorer*
 - File Permissions
 - Security
 - *Different user, different permissions*

that show who can do what with the file

■ No multi-user security for FAT32

different user/permission each user (or group) can have different allowed actions on the same file based on the ACL . if you are logged in as one of the uses you may be allowed to read but not modify , another user may have full control



Windows permissions classifications

■ Full control

- *File can be written to/read from*
- *Permissions can be read and modified*
- *Ownership can be changed*
- *Folder can be listed and entries deleted*

■ Modify

- *Same as Full control*
- *But cannot change permissions or ownership*

Windows permissions classifications

■ Read/Execute

- *File can be read or executed as a program*
- *Folder can be listed and traversed*

■ Read

- *File can be read*
 - But not executed
- *Folder can be listed*
 - But not traversed

■ Write

- *File can be modified*
- *Files/subfolders can be created in a folder*
 - But NOT deleted

■ List folder contents (for folders only)

- *Same as Read/Execute, but not available for files, and only inherited by folders*

Security inheritance

■ Windows:

- *New file or subfolder created, will inherit it's parent's permissions by default*
- *You can override*

■ Unix:

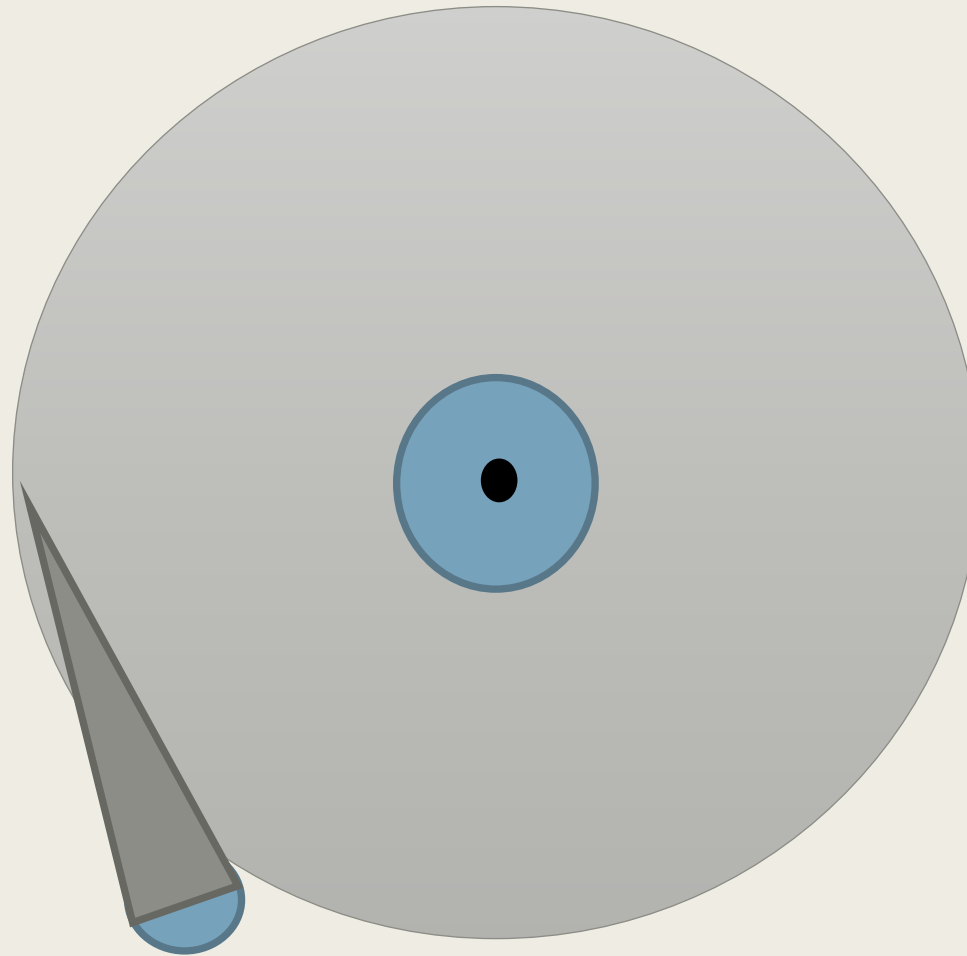
- *Permissions are not inherited for newly created files*
- *Based on user's `umask`* user file creation mask
- *Mask of permissions specific to that user - octal absolute format*

Disk scheduling

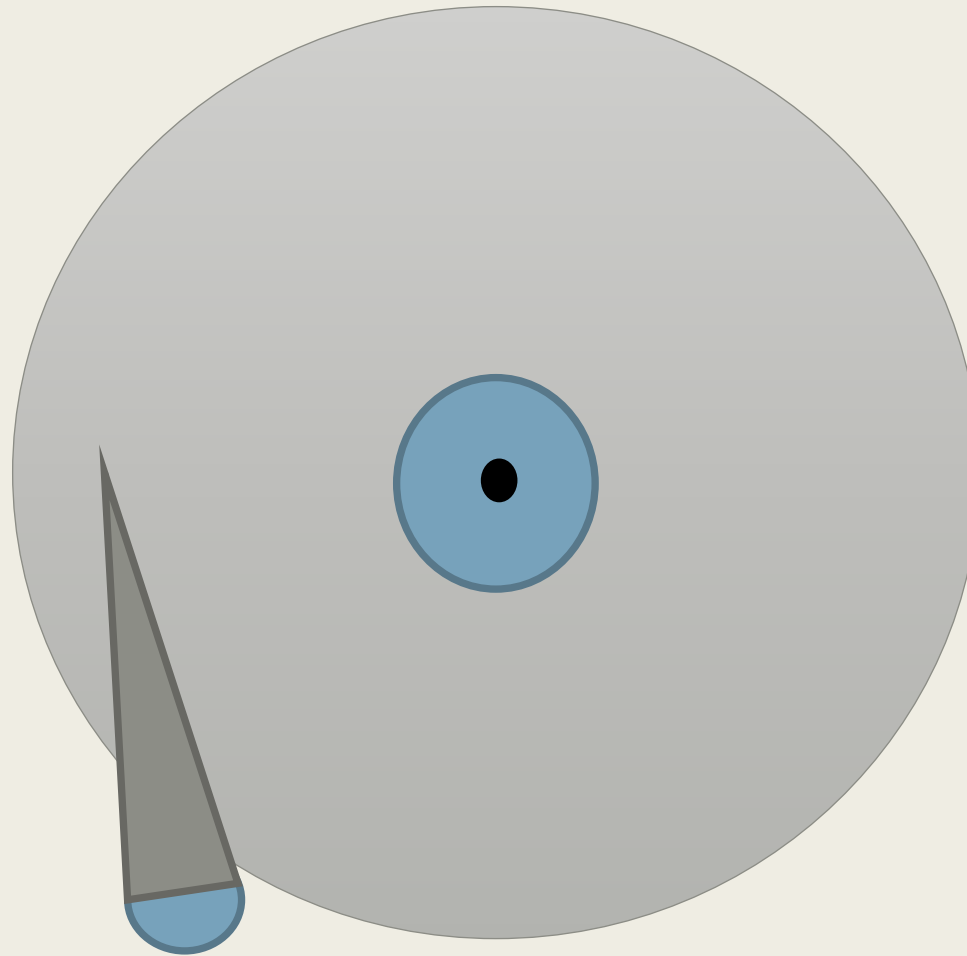
Disk scheduling

- Must be efficient
- Multiple processes, multiple requests to access disk
- Disk scheduling techniques to manage request:
 - ***First-come, first-served (FCFS):** Requests are serviced in the order they arrive, irrespective of positions of heads*
 - ***Shortest-Seek-Time-first (SSTF):** Minimise movement of disk heads*
 - ***SCAN:** Disk heads continuously move in and out, servicing requests as the locations are found.*
 - ***C-SCAN :** Circular scan*
 - ***Look :** Like SCAN, but does not scan all the way to edge*
 - ***C-Look :** Like C-SCAN*

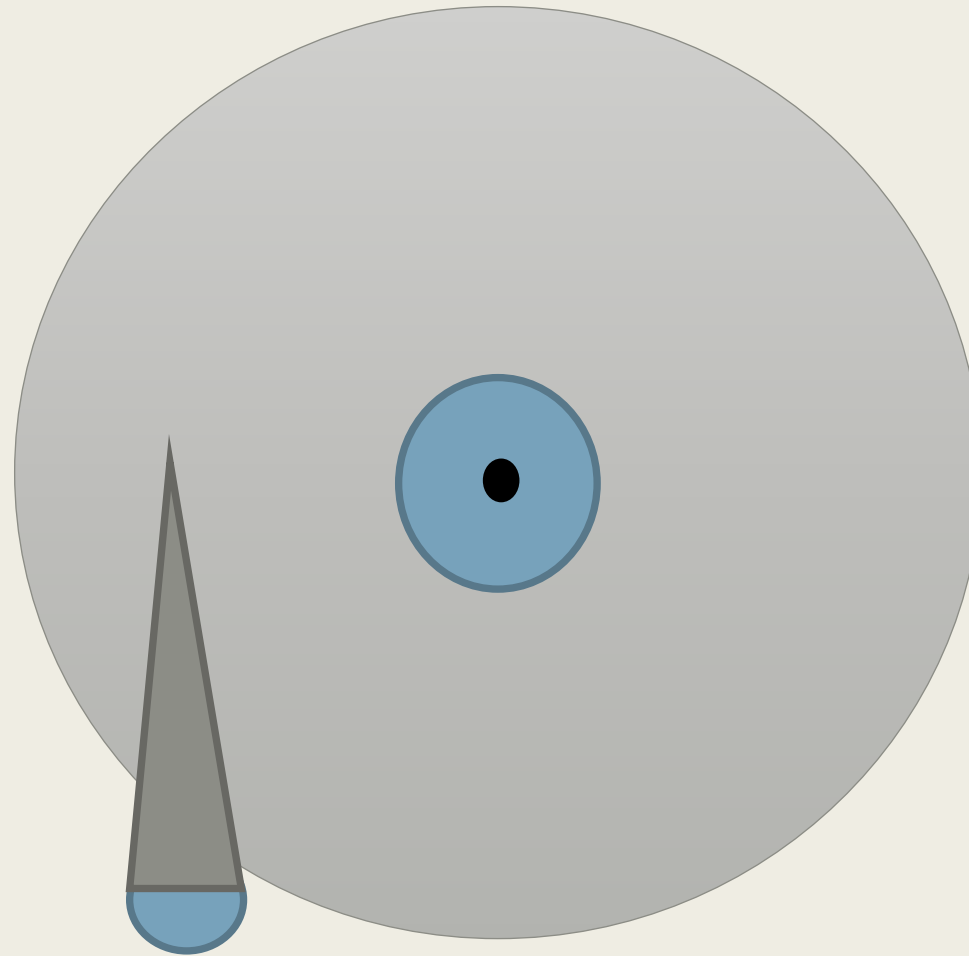
Remember Anatomy of a disk



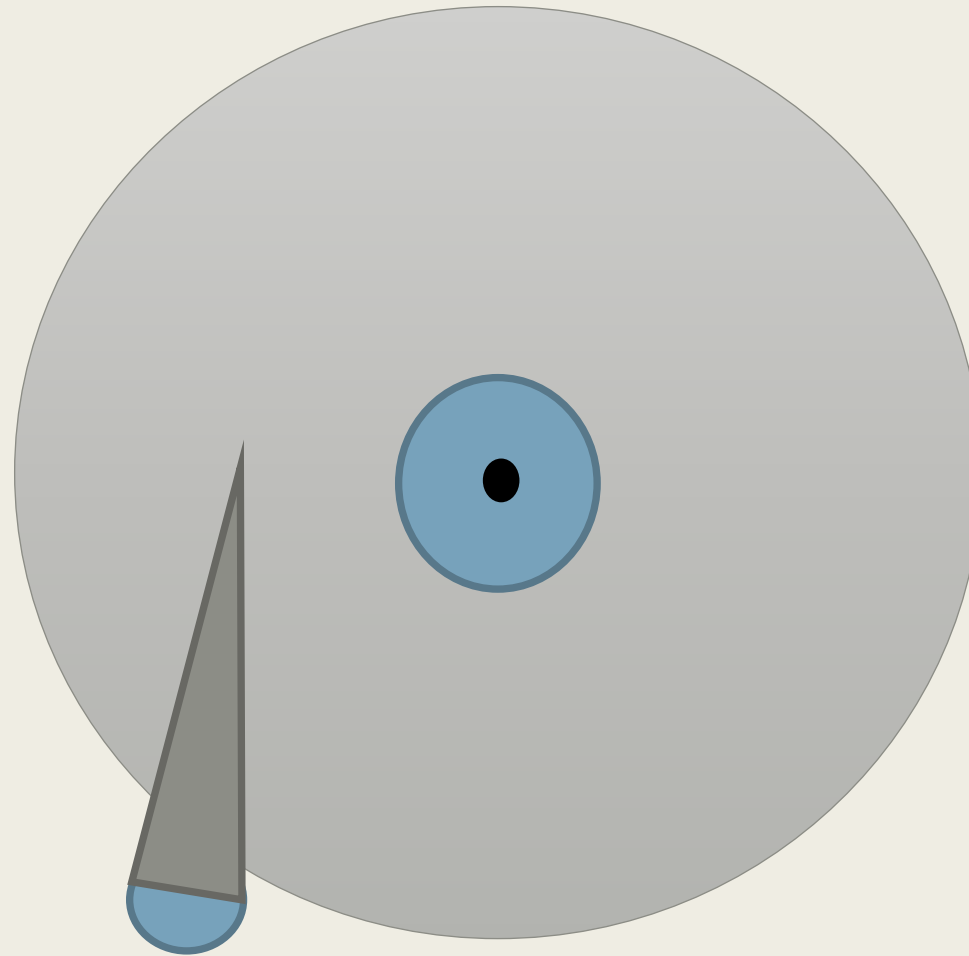
Remember Anatomy of a disk



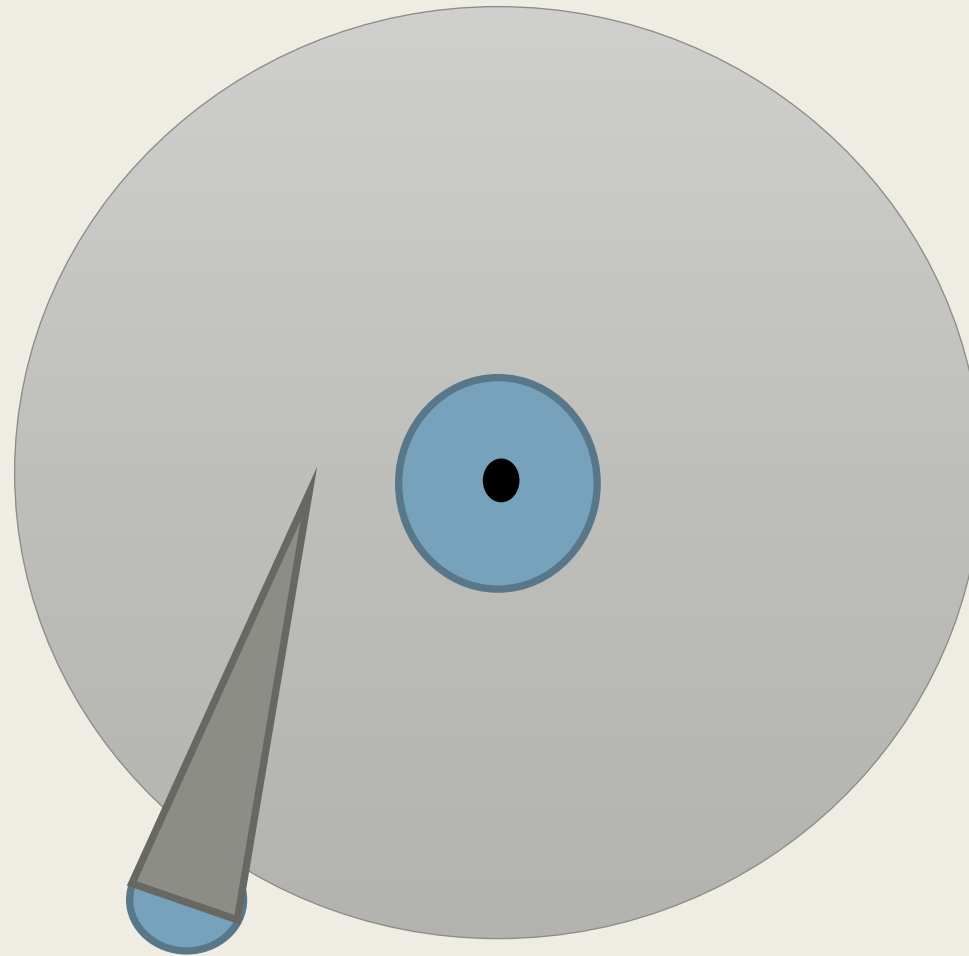
Remember Anatomy of a disk



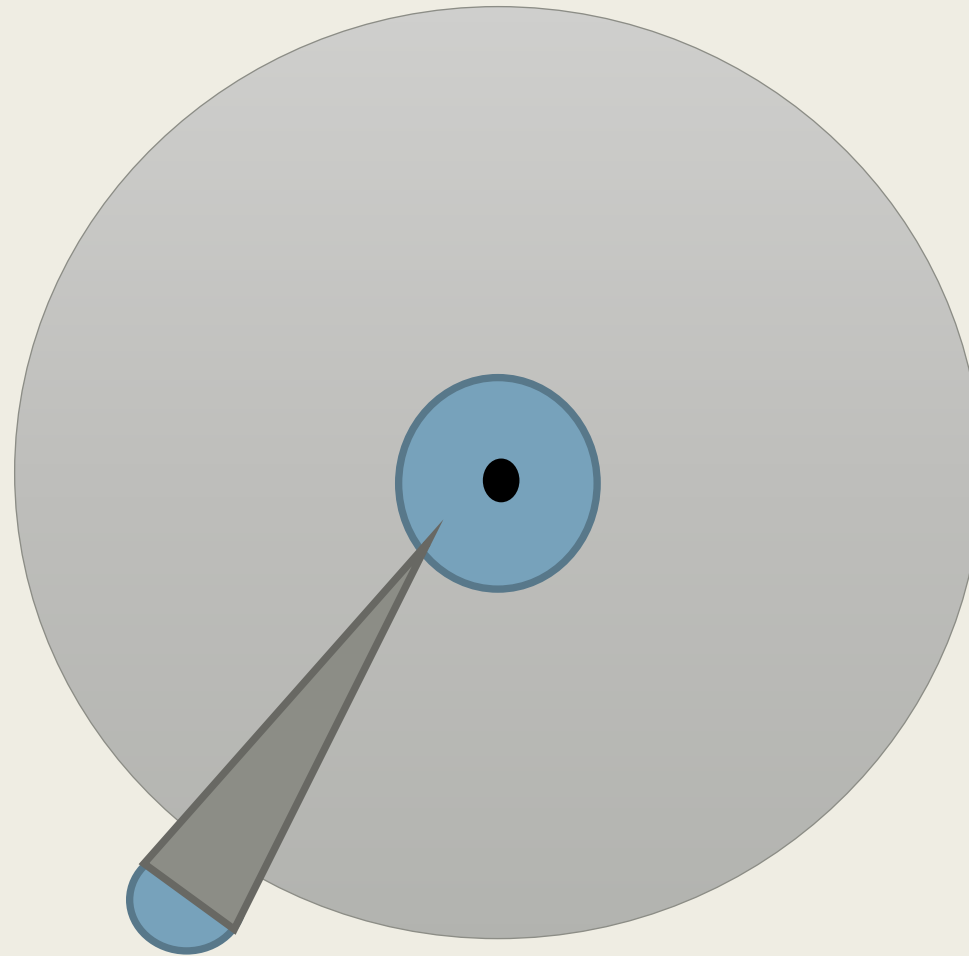
Remember Anatomy of a disk



Remember Anatomy of a disk



Remember Anatomy of a disk



Example:

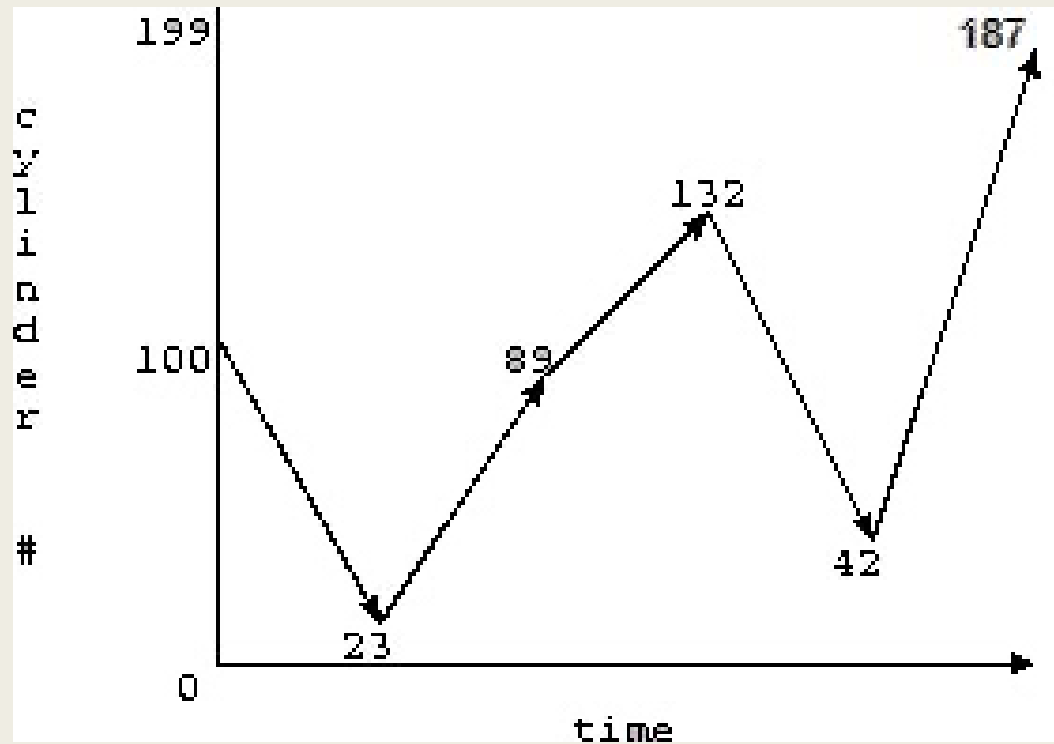
- Work queue: 23, 89, 132, 42, 187
- There are 200 cylinders 0-199
- The disk head starts at number 100

First-Come-First-Served:

■ Work queue: 23, 89, 132, 42, 187

■ Total seek length:

$$|23-100| + |89-23| + |132-89| + |132-89| + |42-132| + |187-42| = 421$$

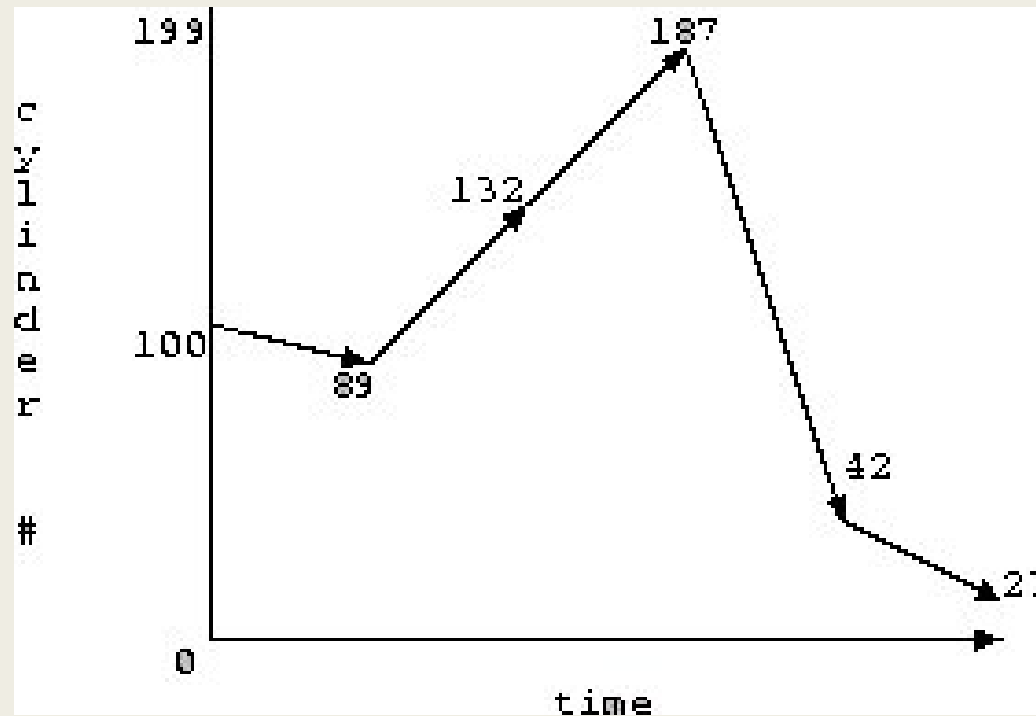


Shortest-Seek-Time-First:

■ Work queue: 23, 89, 132, 42, 187

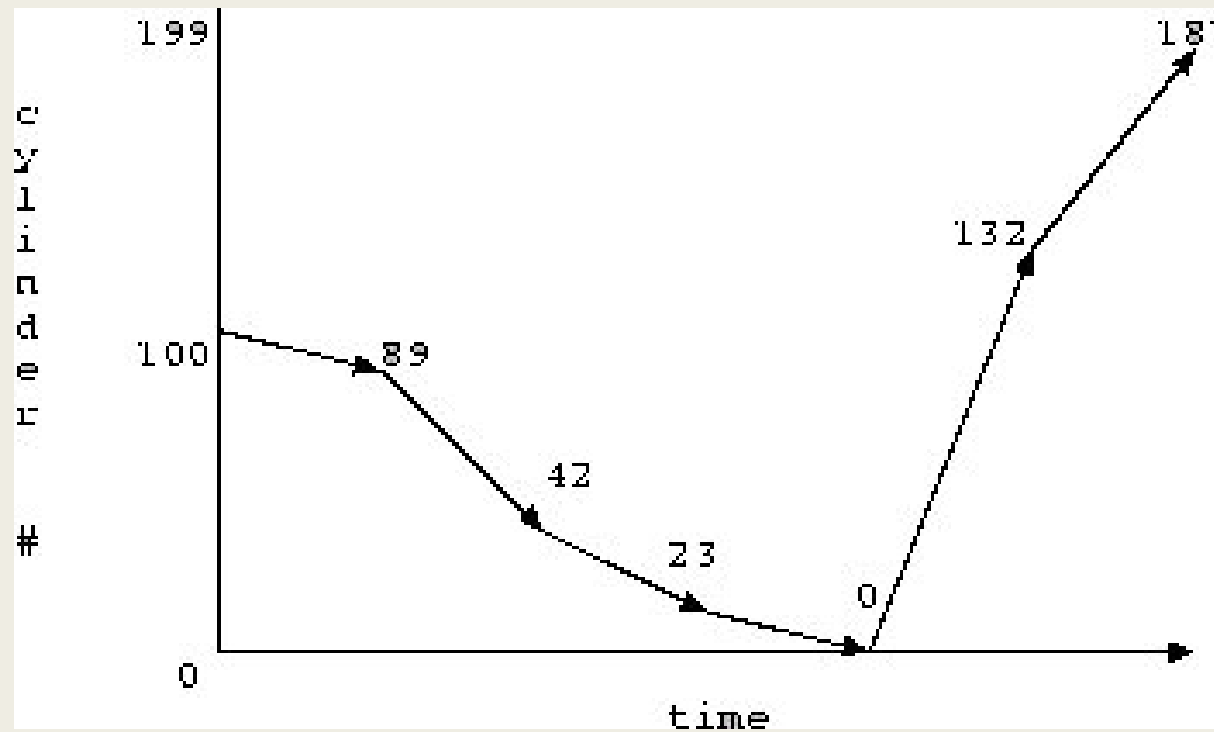
■ Can be inefficient

- *Multiple changing directions*
- *Starvation*



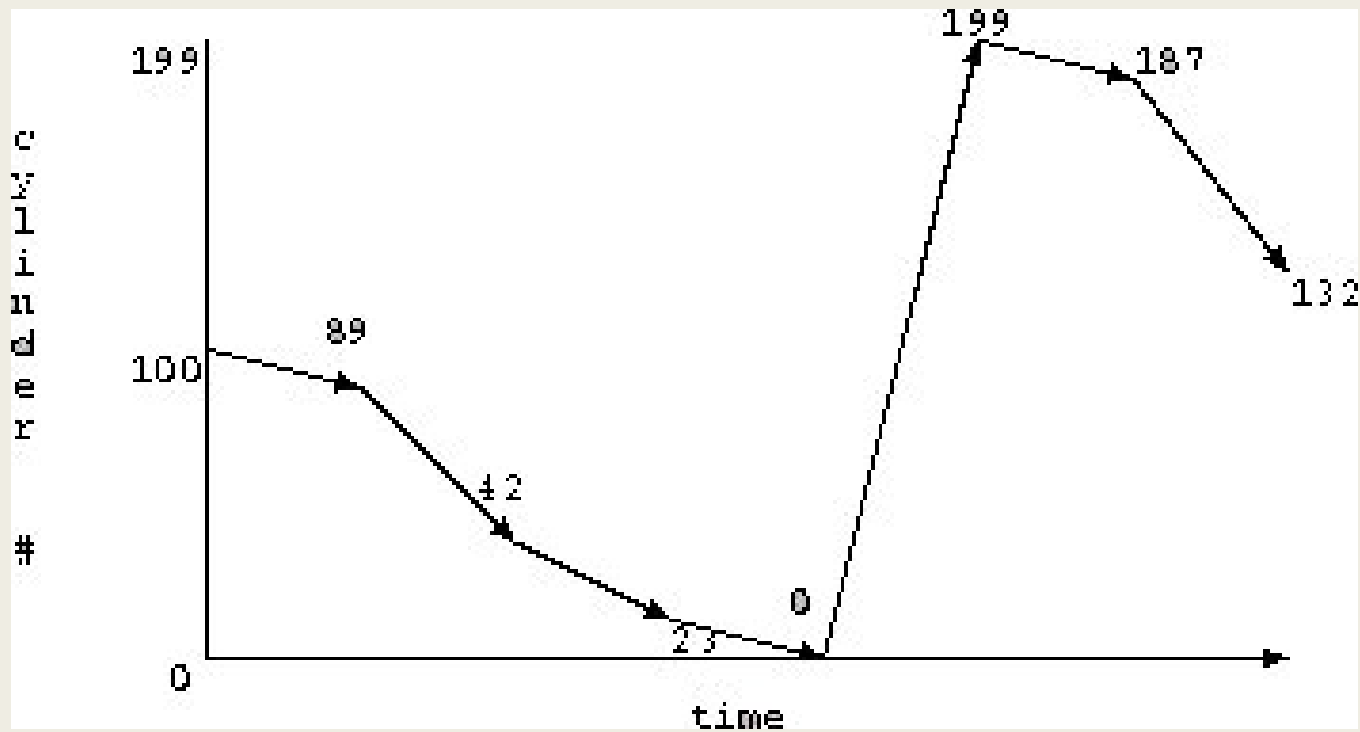
SCAN:

- Work queue: 23, 89, 132, 42, 187
- Elevator
- Sweeps the disk, to-and-fro
- LOOK is similar



C-SCAN:

- Work queue: 23, 89, 132, 42, 187
- Elevator
- Sweeps the disk, but one-direction



Performance:

- Depends on number of requests
- SCAN & C-SCAN are good for systems that place a heavy load on the disk, less likely to cause starvation
- Default: SSTF or LOOK
 - *PRIORITY*

What we covered:

- Partition Sector Zero
 - *FAT16*
 - *Reading & understanding*
- Volumes & Partitions
- Windows & Unix partitions
- Directory structures & terminology
 - *Root Directory*
 - *Parent Directory & Subdirectory*
 - *Absolute Path Names & Relative Path Names*
- File types
- File operations
- Unix file systems
- Disk scheduling

Further reading:

- File signatures reference:

- <https://filesignatures.net/>
- http://www.garykessler.net/library/file_sigs.html

- Indexing & Disk scheduling:

- *Operating Systems: Internals & Design Principles, Williams Stalling (7th ed.)*
 - PP. 550-552 & 510-512
 - Online, see Reading List

Tutorial exercise:

■ File Types:

- *How to establish exactly what the file type of a file is*
- *Even if the extension is wrong*

Questions?

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